

Question #: 1

A 62-year-old woman is undergoing a retrograde pyelogram procedure (injection of contrast into the urethra to visualize the bladder, ureter, and calyces) during the placement of a ureteral stent. During the procedure, a small calculus (stone) is found in one of the minor calyces. Which of the following parts of the kidney most likely projects into each minor calyx?

- A. Renal column
- ✓B. Papilla of pyramid
- C. Cortex
- D. Base of pyramid
- E. Renal sinus

Rationale: The renal papilla is the apex of the renal pyramid that projects into the minor calyx.

Question #: 2

A 24-year-old man is brought to the emergency department following a car crash. Imaging studies show a fracture of the twelfth rib on the left side which has penetrated the kidney to the renal cortex. Which of the following fascial layers is most likely the last to be punctured before the rib entered the kidney?

- A. Perirenal fascia
- ✓B. Renal capsule
- C. Pararenal fascia
- D. Transversalis fascia
- E. Parietal peritoneum

Rationale: The renal capsule is closest to and directly surrounds the renal cortex.

Question #: 3

A 15-year-old boy is brought to the physician by his mother because of a 1-hour history of severe pain after receiving a blow to his groin while playing rugby. Physical examination shows his left testis is swollen and very tender. The physician suspects that the blow has resulted in venous thrombosis resulting in reduced blood flow through the left gonadal veins. The affected veins most likely drains into which of the following veins?

- A. Inferior vena cava

- B. External iliac
- C. Internal iliac
- D. Inferior epigastric
- ✓E. Renal

Rationale: The left gonadal veins drain into the left renal vein, which then drains into the IVC. The right gonadal vein drains directly in the IVC.

Question #: 4

A 60-year-old woman comes to the physician because of a 2-week history of blood in her urinary. Imaging studies show a malignant tumor in the wall of the bladder. Which initial group of lymph node will the tumor most likely drain to?

- A. Parasternal
- B. Deep inguinal
- C. Lumbar
- ✓D. Internal iliac
- E. Superficial inguinal

Rationale: The bladder drains mainly into the internal iliac lymph nodes. It receives blood supply from the internal iliac artery via the superior and inferior vesical arteries. Some lymph from the superior aspect of the bladder drains into the external iliac nodes.

Question #: 5

A 44-year-old man comes to the emergency department because of excessive weight loss, left sided flank pain and hematuria. Physical examination shows pale mucous membranes and a left side intra-abdominal mass. CT scan shows a large multilocular mass that has within the parenchyma of the kidney. Histological examination shows a renal cell carcinoma which originated in the distal convoluted tubules. Which of the following embryonic structures is the source of the precursor cells of this cancer?

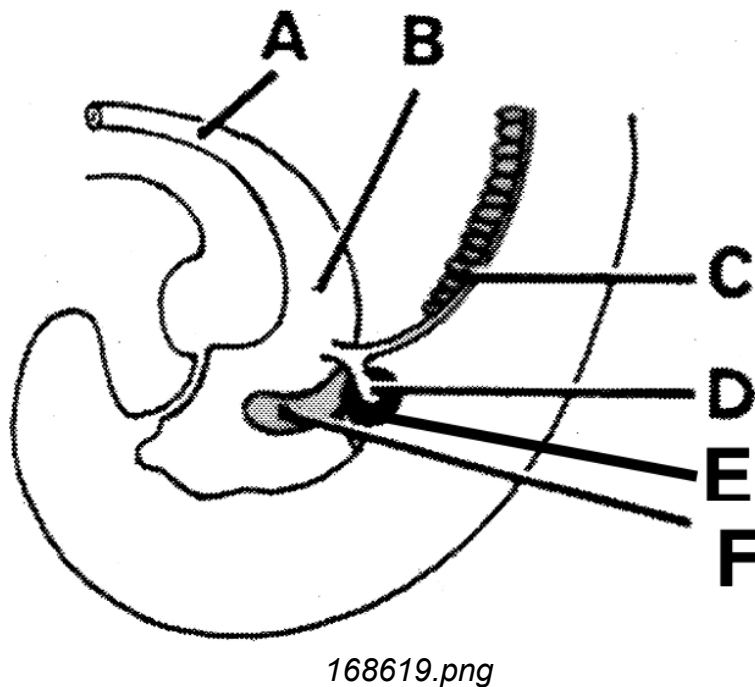
- A. A
- B. B
- C. C
- D. D
- ✓E. E

Rationale: Correct answer: Excretory part – Bowman's capsule, Proximal Convoluted Tubules (PCT), loop of Henle and Distal Convoluted Tubules (DCT) - Develop from mesenchyme of the metanephric blastema.

Incorrect answer: Structure labeled D: Conducting part - Collecting tubules, minor calyces, major calyces, pelvis, and ureter - develop from the ureteric bud.

Structure labelled F: Cloaca
Structure labelled C: Mesonephros

Attachment:



Question #: 6

A 25-year-old man is brought to the emergency department after he was hit in his back with a baseball bat. X-ray of the chest and abdomen shows no bony abnormalities. Ultrasound shows no free collection of fluid in the abdomen however the left renal fossa was empty. He is further evaluated with imaging studies, and the results are represented in the image provided. Which of the following best describes the embryonic origin of the abnormal finding?

- A. Complete early bifurcation of the ureteric bud of the left kidney
- B. Incomplete early bifurcation of the ureteric bud of the right kidney
- C. Fusion of the inferior pole of the kidneys
- D. Failure of ascent of the kidneys
- E. Multiple ureteric buds
- F. Failure of formation of the ureteric buds
- ✓G. Migration and fusion of the left kidney to the right

Rationale: Correct answer: As seen on the Retrograde Ureterogram, there is migration and fusion of the left kidney to the right side. As there are two kidneys, but only one ureter.

Incorrect answers: Some time One ureteric bud could divide and induce the formation of two kidneys, but in that case, we will see two ureters on the effected side. In case of incomplete division of the ureteric bud, there will be one kidney and two ureters. Fusion of the inferior pole

of the kidneys will lead to horseshoe kidney.

Attachment:



168621.png

Question #: 7

A 24-year-old man is brought to the emergency department after being hit by a truck. On physical examination the patient is unconscious and there is bruising of the skin over the left kidney. Vitals: BP 85/65, pulse rate 110, RR 22. CT scan shows a hematoma collecting in the fascia that lies directly external to the renal fascia. Which of the following is the name given to this tissue?

- A. Perirenal fat
- B. Transversalis fascia
- C. Renal capsule
- ✓D. Pararenal fat

E. Parietal peritoneum

Rationale: Correct answer: Outside of the renal fascia is layer of fat tissue called the pararenal/paranephric fat

Incorrect answer: Perirenal (perinephric) fat : Enclosing inside the renal fascia is another layer of fat. Found between the renal fascia and renal capsule.

Question #: 8

A 58-year-old man presents to the emergency department with sudden-onset severe chest pain radiating to his left arm. Electrocardiogram shows ST-segment elevation in the anterior leads. Cardiac catheterization shows occlusion of the left anterior descending coronary artery, resulting in markedly reduced blood flow to the anterior portion of the heart. It is observed that oxygen delivery to the affected myocardium is severely reduced, and the cardiac muscle cannot rely on mitochondrial oxidative phosphorylation to meet its energy demands. Which of the following enzymes is most likely responsible for producing ATP in the ischemic cardiac muscle under these conditions?

- A. Phosphofructokinase-1, phosphoglycerate kinase
- B. Phosphoglycerate kinase, pyruvate dehydrogenase
- C. Phosphoglyceromutase, pyruvate dehydrogenase
- D. Phosphofructokinase-1, pyruvate kinase
- ✓E. Phosphoglycerate kinase, pyruvate kinase

Rationale: Glycolysis provides ATP to tissues when there is hypoxia. Under conditions of low oxygen tension, the substrate level phosphorylation reaction in glycolysis provide (make) ATP for survival of the tissues. Lactate is the end-product under these conditions. The two enzymes that form ATP in glycolysis are phosphoglycerate kinase and pyruvate kinase.

Question #: 9

A 28-year-old woman presents to the physician because of a 2-week history of weakness and fatigue. She reports a history of recurrent episodes of jaundice and pallor since childhood. Physical examination shows mild splenomegaly. Laboratory studies show:

- Hemoglobin: 8.5 g/dL (low)
- Reticulocyte count: elevated
- Lactate dehydrogenase: elevated
- Haptoglobin: decreased
- Direct bilirubin: normal; indirect bilirubin: elevated

Erythrocyte enzyme analysis shows 20% activity of pyruvate kinase. Which of the following is

the most likely mechanism of hemolysis in this patient?

- A. Increased accumulation of reactive oxygen species
- B. Increased levels of 2,3-bisphosphoglycerate
- C. Accumulation of alpha-globin chains in red blood cells
- ✓D. Decreased activity of erythrocyte Na⁺-K⁺ ATPase
- E. Polymerization of hemoglobin
- F. Imbalance in the alpha globin to beta globin ratio

Rationale: The vignette describes a patient with hemolytic anemia due to pyruvate kinase deficiency. Pyruvate kinase is an enzyme of glycolysis, and glycolysis aims to provide energy to RBCs for the functioning of the various active transporters (Na-K ATPase). Glycolytic defects result in reduced ATP generation and impaired functioning of the pumps to maintain osmotic balance in cells. The RBCs contain higher levels of 2,3-BPG in pyruvate kinase deficiency, which is NOT responsible for hemolysis.

Hemolysis in G6PD deficiency is due to increased accumulation of reactive oxygen species due to impaired formation of NADPH required to maintain glutathione in the reduced state.

In beta-thalassemia, there is an accumulation of alpha-globin chains, which results in hemolysis. In HbS disease, hemoglobin polymerization results in distorted crescent-shaped RBCs and hemolytic anemia.

In alpha and beta thalassemia, there is an imbalance in the alpha-to-beta globin ratio.

Question #: 10

A 6-month-old boy is brought to the physician by his parents because of a 1-month history of pallor. His past medical history shows anemia. Laboratory studies show:

- Hemoglobin: 6.5 g/dL (low)
- Reticulocyte count: elevated
- Lactate dehydrogenase: elevated
- Haptoglobin: decreased

Hemoglobin electrophoresis shows a predominance of normal hemoglobin. The oxygen dissociation curve of the patient's hemoglobin (red line) is compared with that of normal controls (black line), as shown. Which of the following metabolic changes in the patient's erythrocytes is most likely responsible for the observed shift in oxygen binding?

- ✓A. Increased 2,3-bisphosphoglycerate levels
- B. Increased HbA1c levels
- C. Increased arterial pH
- D. Decreased ATP levels
- E. Decreased NADPH levels

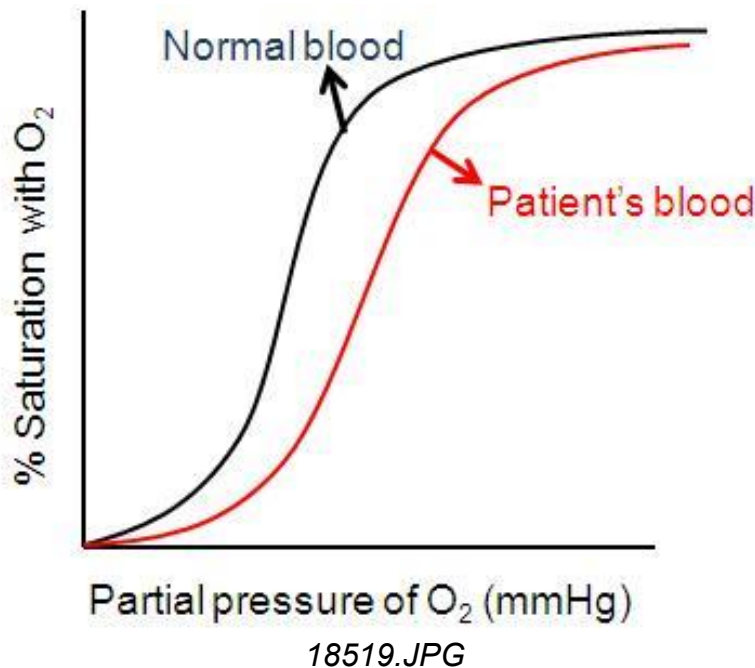
F. Increased HbF levels

G. Increased reactive oxygen species

Rationale: The vignette describes a patient with hemolytic anemia due to pyruvate kinase deficiency. Note that there is a right shift of the hemoglobin oxygen dissociation curve. This is due to elevated 2,3-BPG levels in RBCs in pyruvate kinase deficiency. As a result of elevated 2,3-BPG levels, there is decreased oxygen affinity and hence more oxygen unloading. Patients with higher 2,3-BPG levels tend to have less severe presentations.

Note that HbA1c (glycated hemoglobin), NADPH, and reactive oxygen species do not affect the Hb oxygen dissociation curve. Also, these parameters are not altered in a patient with pyruvate kinase deficiency.

Attachment:



Question #: 11

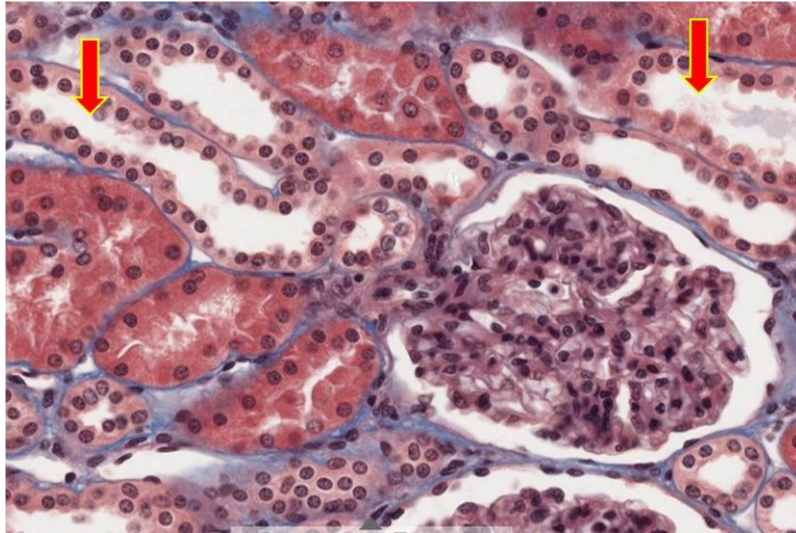
A 30-year-old man comes to the physician because of a 2-day history of puffiness in the face and feet, associated with headache and palpitation. He has a family history of several first-order relatives having essential hypertension. Physical examination shows pitting edema and palor. His pulse is 111/min, blood pressure is 190/100 mm Hg, and temperature is 37.5°C (99.5°F). Laboratory studies show signs of renal failure. A renal ultrasound shows shrunken kidneys with multiple cystic structures. Examination of a biopsy specimen of the kidneys show dilatation of renal tubules and very little normal renal parenchyma. The attached photomicrograph shows healthy regions of the affected tissue. Which of the following best describes the tubular structures indicated by the red pointers?

- A. Continuation of Bowman's capsule
- B. Re-absorption of glucose
- ✓C. Joins collecting duct via connecting tubule

- D. Production and secretion of renin
- E. Component of the renal corpuscle

Rationale: The arrows are pointing to the distal convoluted tubules of the pars convoluta of the cortical labyrinth. DCT has epithelial cells that are much smaller in comparison to the proximal convoluted tubule, as well as a "cleaner" lumen.

Attachment:



Picture14.jpg

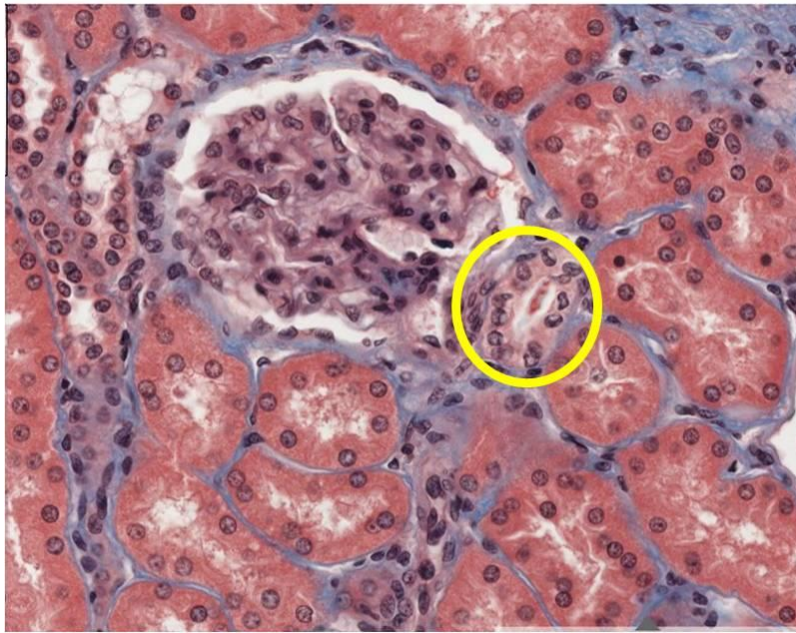
Question #: 12

A 62-year-old woman comes to the physician because of a 3-month history of swelling of both legs. Physical examination shows bilateral leg edema. A kidney biopsy is collected for histopathological examination and shown in the attached photomicrograph. Which of the following statements best characterizes the indicated structure?

- A. Functions in reabsorption of amino acids from filtrate
- B. Functions in reabsorption of glucose from filtrate
- ✓C. Monitors the sodium content of the filtrate
- D. Has abundant aquaporins for the secretion of water
- E. Delivers filtrate to the minor calyx

Rationale: The macula densa consists of a group of 15-20 modified cells that appear narrower and taller than other distal tubule cells. These cells function to sense changes in the salt content of the tubular fluid.

Attachment:



MD.jpg

Question #: 13

A 30-year-old man is brought to the emergency department because of a 30-minute history of collapse following involvement in a car crash. Sudden fall in blood pressure is stabilized by reabsorption of sodium and water by the kidney. Renin, which initiates this process, is most likely produced by which of the following?

- A. Extraglomerular mesangial cells
- B. Intraglomerular mesangial cells
- C. Macula densa
- D. Endothelial cells of glomerular capillaries
- ✓E. Juxtaglomerular cells

Rationale: Renin is produced by the juxtaglomerular cells which are modified cells of the afferent arteriole that form part of the juxtaglomerular apparatus that regulates blood pressure. Renin in turn cleaves angiotensinogen (produced by the liver) to angiotensin I.

Question #: 14

A 23-year-old man comes to the physician because of a 1-week history of reddish urine and cough with expectoration of bloody sputum. Physical examination shows generalized edema. Laboratory investigations showed antibodies against the glomerular basement membrane (GBM). Examination of biopsy specimens show deposits of antibodies on the GBM as well as crescent-shaped inflammatory infiltrates within the renal corpuscles. Which of the following regions of the glomerular filtration barrier in the attached photomicrograph is the most likely

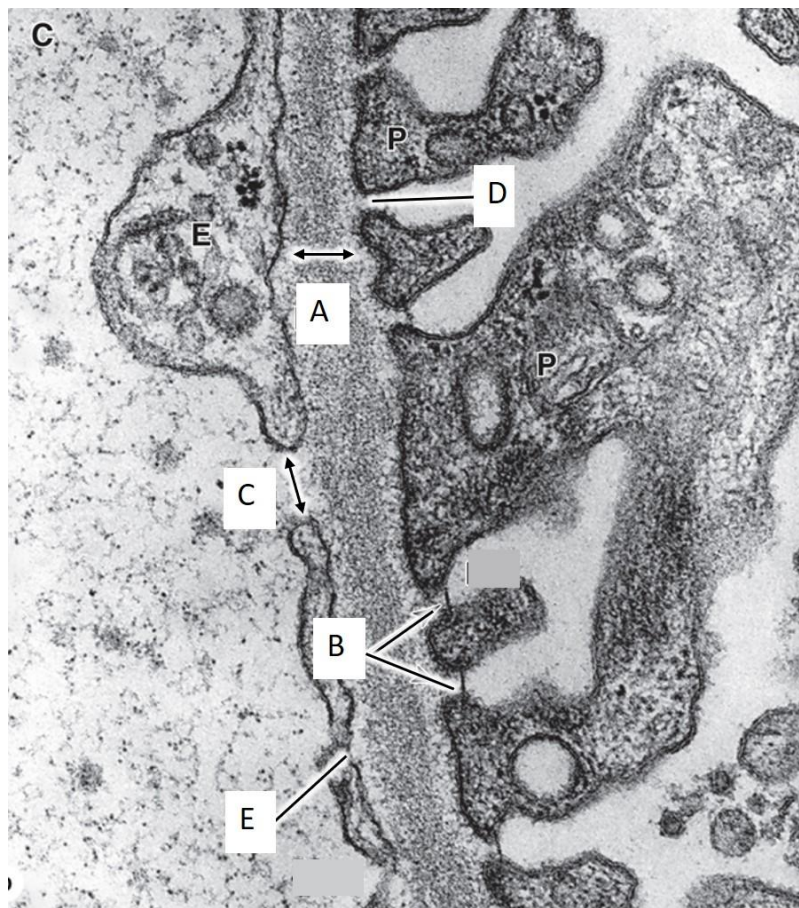
target for the antibodies?

- ✓A. A
- B. B
- C. C
- D. D
- E. E

Rationale: The Glomerular basement membrane being targeted by the antibodies is depicted by the option (A). It restricts the movement of particles larger than 70KDa. Antibody-mediated damage to this membrane is responsible for the reddish-tinged urine. The patient has Goodpasture syndrome in which antibodies attack the type IV collagen in glomerular basement membrane in the kidney and respiratory basement membrane in the lungs, hence his additional symptom of cough with bloody sputum.

(B) is labeling the filtration slit diaphragms and (D) is labeling the filtration slits themselves. (E) denotes glomerular endothelial cells. (C) denotes glomerular endothelial fenestrations.

Attachment:



Glomerular filtration barrier.jpg

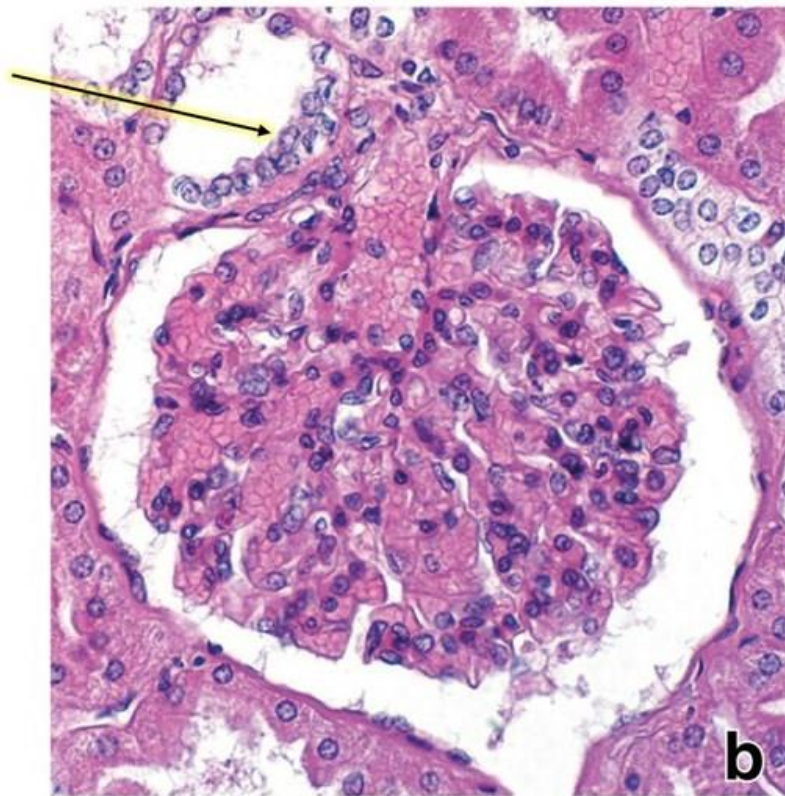
A 65-year-old man comes to the emergency department because of a 2-month history of lightheadedness, dizziness, and malaise. His blood pressure is 80/50 mm Hg, and pulse is 110/min. Physical examination shows weak peripheral pulses with pale and cold distal extremities. The changes in the urinary filtrate brought about by the decreased blood pressure are monitored by the cells shown in the attached photomicrograph. Which of the following statements best characterizes these cells?

- A. Endothelial cells of afferent arterioles
- ✓B. Epithelial cells of the distal straight tubule
- C. Epithelial cells of the proximal tubule
- D. Extraglomerular mesangial cells
- E. Modified smooth muscle cells of the afferent arteriole

Rationale: The cells being indicated by the pointer are the Macula Densa cells. They are a population of tall columnar cells of the distal straight tubule.

They monitor the tubular concentration of Na^+ . They are also a part of the juxtaglomerular apparatus and play a role in the Renin-Angiotensin-Aldosterone system.

Attachment:



Renal Corpuscle.jpg

Question #: 16

A 32-year-old man is resting under normal atmospheric conditions. Due to his respiratory response to PaCO_2 , his central chemoreceptors are stimulated. Which of the following best

describes the main stimulating factor and site of location for these receptors?

- A. Carbon dioxide in the lung alveoli
- ✓B. Hydrogen ion on the ventral surface of the medulla
- C. Hydrogen ion in the aortic arch
- D. Carbon dioxide in the carotid bodies
- E. Carbon dioxide in the chemosensitive area of the medulla

Rationale: The main stimulating factor for the central chemoreceptors are the H^+ increase, which is an indirect reflection of increased arterial PCO_2 , so the correct answer is Hydrogen ion on the ventral surface of the medulla

Question #: 17

A 61-year-old woman comes to the physician because of a 5-day history of shortness of breath. She had an inferior wall myocardial infarction 2-years ago. Her pulse is 88/min, blood pressure is 128/84 mmHg, respirations are 26/min, and oxygen saturation is 86% in room air. Physical examination shows bilateral basal rales. She is diagnosed with acute pulmonary edema. Which of the following is the most likely cause of hypoxemia in this patient?

- A. Retention of CO_2
- B. Left-shifted oxy-hemoglobin dissociation curve
- C. V/Q mismatch
- ✓D. Increased thickness of the respiratory membrane
- E. Decreased A-a gradient

Rationale: This question is asking about why oxygen diffusion is defective in pulmonary edema. In pulmonary edema, accumulation of fluid in the interstitial space increases the thickness of the alveolocapillary membrane across which diffusion takes place. According to Fick's law of diffusion, if the thickness of membrane increases, the diffusion of gas suffers. Oxygen behaves as a diffusion-limited gas in this situation.

Question #: 18

A 1-day-old newborn boy is being treated in the neonatal intensive care unit because of respiratory distress. He was born at 28 weeks of gestation. His pulse is 146/min, respirations are 45/min, and blood pressure is 80/50 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 82% (Normal: 95-98%). Which of the following lung alterations is most likely associated with this patient's condition?

- A. Increased work on alveolar deflation
- B. Increased surfactant

- C. Emphysema
- ✓D. Increased work on alveolar distension
- E. Decreased surface tension

Rationale: RDS (respiratory distress syndrome) is a syndrome of respiratory difficulty in newborn infants caused by a deficiency of surfactant. The air-fluid interface of the film of water lining the alveoli of the lung (where the exchange of oxygen and CO₂ occurs) exerts large forces that cause the alveoli to close if surfactant is deficient. Lung compliance has decreased, and the work of inflating the stiff lungs is increased. The preterm newborn is further handicapped because his or her ribs are more easily deformed (compliant). Breathing efforts therefore result in deep sternal (breastbone) retractions but poor air entry if the ribs are compliant compared with the lungs. This results in diffuse atelectasis (collapse of the lungs).

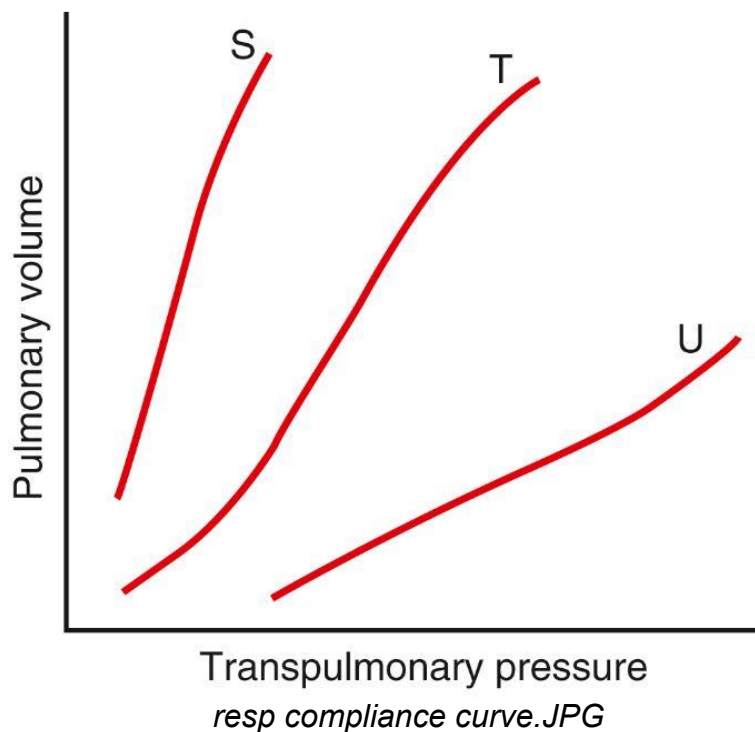
Question #: 19

A study is conducted to determine the compliance response of isolated lungs when exposed to various transpulmonary pressures. The changes in pulmonary volume are plotted against the changes in transpulmonary pressure. The graph in the attached image shows three different compliance curves (S, T and U). Which of the following best describes the relative compliances for these three curves?

- A. $S < T < U$
- B. $S < T > U$
- C. $S = T = U$
- D. $S > T < U$
- ✓E. $S > T > U$

Rationale: Compliance is the change in lung volume that occurs for a given change in the transpulmonary pressure. Compliance is equal to the slope of the volume-pressure relationship. S is greatest compliance. U is the lowest compliance.

Attachment:



Question #: 20

A 55-year-old man is brought to the emergency department because of a 2-hour history of severe difficulty breathing. His wife says he has fever, runny nose, and loss of sense of smell during the past 5 days. His temperature is 39°C(102°F), pulse is 95/min, respirations are 22/min, and blood pressure is 130/83 mm Hg. Physical examination shows bluish lips and fingernails and use of accessory muscles of breathing. He is diagnosed with acute respiratory distress because of lung surfactant deficiency. Which of the following best explains the normal function of the lung surfactants?

- A. Decrease surface tension in the larger alveoli
- B. Increase surface tension in the smaller alveoli
- C. Increase fluid movement into the interstitial space
- D. Increase hydrostatic pressure in pulmonary capillaries
- ✓E. Prevent overdistension of larger alveoli and collapse of small alveoli

Rationale: Surfactant is important in stabilizing alveoli in the lungs. Surfactants decrease surface tension when they are spread in a smaller surface area. In larger alveoli, they are spread apart and increase surface tension. Thus, by altering the surface tension depending on the size of the alveoli, they stabilize the alveoli. This results in prevention of overdistension of larger alveoli and collapse of smaller alveoli.

Question #: 21

A 56-year-old woman comes to the physician because of a 10-month history of progressive shortness of breath. She was diagnosed with rheumatoid arthritis 4 years ago. Her pulse is 92/min, respirations are 22/min, and blood pressure is 134/85 mmHg. Physical examination shows deformities in small joints of her hands. Results of the pulmonary function studies show evidence of restrictive lung disease. Which of the following changes in the lung volumes/capacities is most likely in this patient?

- A. Increased tidal volume and FRC
- ✓B. Decreased RV and FRC
- C. Increased RV and decreased FRC
- D. Decreased RV and increased FRC
- E. Increased tidal volume and FEV₁

Rationale: patients with restrictive lung diseases will have a small lung with decreased RV, FRC, tidal volume and TLC.

Question #: 22

A 30-year-old man wants to run in a marathon for the first time. Prior to the race his pulse is 60/min, breathing is 14/min, O₂ saturation is 98%. He decides to inhale 100% oxygen one day before the race to enhance his performance after seeing an advertisement from an oxygen bar. This therapy would most likely result in which of the following outcomes?

- A. Increases O₂ carried by Hb, but dissolved O₂ will be the same
- ✓B. Will not increase the total O₂ content appreciably
- C. The O₂ content will rise 5-fold and enhance his performance greatly
- D. A-a gradient increases for a week
- E. PaO₂ increases during running

Rationale: Oxygen carried by Hb will not increase because the saturation of hemoglobin is almost full (100%) in this person and further binding of O₂ is not possible even if the partial pressure increases. As total oxygen content depends on oxygen bound to Hb, the total O₂ content will not increase appreciably. A-a gradient will not increase because there is no gross difference between PAO₂ and PaO₂. PaO₂ may increase when he inhales 100% O₂, however, it is a transient change that comes back to normal after stopping this inhalation.

Question #: 23

A 25-year-old pregnant woman comes the physician for a routine examination. Her vital signs are within normal limits and physical examination shows no abnormalities. She is advised to lie on her left side when resting. Which of the following is most likely in the left lung as compared to the right lung when the patient is in this position?

- A. Larger alveoli will be present in the left lung
- B. A higher V/Q ratio in the left lung
- ✓C. Greater ventilation per unit volume in left lung
- D. In left lung lower blood flow per unit volume
- E. Lower PaCO₂ in left lung

Rationale: In this position the left side of the lung is at a lower level and the right side of the lung is at an upper level (similar to base and apex of a normal lung, respectively). The alveoli at the base (left side) will be smaller and more compliant than those at the apex (right side) because of the effect of gravity. More compliant alveoli has the potential to facilitate higher ventilation.

At the base, perfusion is in excess of ventilation because of gravity. The opposite is true for the apex. Therefore, the right side of the lung will have less blood flow than the left.

Ventilation is in excess of perfusion at the apex, while the opposite is true for the base. There is therefore more exchange of CO₂ from blood to the alveoli at the apex than at the base.

Question #: 24

A 25-year-old man comes to the physician for a routine examination. His vital signs are within normal limits. Physical examination shows no abnormalities. Physician asks him to lie down on his left side in a relaxed state breathing normally. Which of the following would be mostly expected in the right lung as compared with the left lung?

- A. Lesser ventilation per unit volume in the left lung
- B. A higher V/Q ratio in the left lung
- C. Greater ventilation per unit volume in the right lung
- D. Lower blood flow per unit volume in the left lung
- ✓E. A higher V/Q ratio in the right lung

Rationale: In this position, the left lung is lower (closer to gravity), and the right lung is upper (away from gravity). Because of this, the left lung has greater blood flow and ventilation compared to the right lung. In the right lung, both ventilation and blood flow are reduced, with blood flow reduction more than that of ventilation. Therefore, the V/Q ratio is higher in the right lung.

Question #: 25

A 45-year-old man comes to the physician for a routine examination. He says he has difficulty breathing when he goes snorkeling. Results of the spirometry show:

Tidal volume = 550 mL

Anatomical dead space = 125 mL

Respiratory rate = 15/min

FEV₁/FVC ratio = 82%.

The snorkeling gear that he uses has a volume of 90 mL. Which of the following values best approximates this patient's alveolar ventilation whilst breathing through the snorkeling gear if he

breathes with the above tidal volume and respiratory rate?

- ✓A. 5025 mL/min
- B. 6600 ml/min
- C. 7200 mL/min
- D. 7800 mL/min
- E. 8500 mL/min

Rationale: $VA = (VT - VD) \times F$

$$VD = 125 + 90$$

$$VA = (550 - 215) = 335 \times 15$$

$$VA = 5025 \text{ ml/min}$$

The snorkel is a breathing tube used for breathing when swimming with your face underwater. This tube increases the anatomical dead space; therefore, it must be considered in the assessment of the alveolar ventilation of this patient.

Question #: 26

An 86-year-old man comes to the emergency department because of a 1-week history of difficulty breathing. He has a 15-year history of chronic obstructive pulmonary disease. Results of arterial blood gas studies show:

Arterial O₂ (P_{aO2}) = 55 mm Hg

Arterial O₂ saturation = 78%

Arterial CO₂ (P_{aCO2}) = 52 mm Hg

pH = 7.31.

Assume atmospheric pressure (P_{atm}) = 760 mm Hg, P_{H2O}) = 47 mm Hg, and Respiratory quotient (R) = 0.8.

Which of the following pressures best approximates alveolar PO₂ (P_{AO2}) in this patient?

- A. 55 mm Hg
- B. 64 mm Hg
- C. 78 mm Hg
- ✓D. 85 mm Hg
- E. 102 mm Hg

Rationale: Use alveolar gas equation to calculate PAO₂.

$$\begin{aligned} PAO_2 &= (FiO_2 \times (P_{atm} - P_{H_2O})) - (P_{aCO_2} / RQ). \\ &= 21\% \times (760 - 47) - (52 / 0.8) \\ &= 149.7 - 65 \\ &= 84.7 \text{ mm Hg} \end{aligned}$$

Question #: 27

A 64-year-old woman is brought to the emergency department because of a 2-day history of fever and a 2-hour history of shortness of breath. Her pulse is 120/min. Physical examination shows rapid movements of the thoracic cage with increased respiratory rate but reduced tidal volume during breathing. Laboratory studies show low PaO₂, low oxygen saturation, and hypoxemia. Which of the following clinical features are most likely seen in this patient?

1. Shallow breathing
2. Deep Breathing
3. Tachypnea
4. Bradypnea
5. Tachycardia
6. Bradycardia

- ✓A. 1, 3, 5
B. 2, 4, 6
C. 1, 4, 6
D. 2, 3, 5
E. 1, 4, 5

Rationale: Shallow breathing, Tachypnea, and Tachycardia are the most likely features. Hypoxemia stimulated respiratory center. This has resulted in increased rate of breathing (tachypnea) but with reduced tidal volume (shallow breathing). The heart rate is high (tachycardia) to increase stroke volume, to keep the extra effort for breathing.

Question #: 28

A 21-year-old man comes to the physician because of a 2-week history of swelling on the right side of the neck. He says the swelling is not painful. He has a family history of Hodgkin's disease. He appears anxious. His respirations are 24/min with increased tidal volume; other vital signs are within normal limits. Physical examination shows palpable, non-tender anterior cervical lymph nodes. Pulmonary studies show increased alveolar ventilation. Which of the following parameters is most likely decreased in this patient?

- A. PaO₂
B. V/Q ratio
✓C. PaCO₂
D. A-a gradient
E. Minute ventilation

Rationale: The patient is hyperventilating because his breathing rate is increased to 22 /min from the normal 12 /min. This would result in more CO₂ being exhaled, decreasing the arterial partial pressure of CO₂ (PaCO₂), making this the correct answer. Blood pH may also increase (respiratory alkalosis). Minute ventilation would be increased, and V/Q ratio and PaO₂ may increase due to the elevated ventilation. A-a ratio is unlikely to change as a result of the

hyperventilation.

Question #: 29

A 32-year-old man comes to the physician because of a 3-week history of difficulty breathing. His respiratory rate is 12/min and pulse rate is 78/min. A pulmonary function test shows the following:

Tidal volume = 680 mL

Anatomic dead space = 150 mL

Physiologic dead space = 180 mL

Which of the following values best approximates his alveolar ventilation?

A. 4500 mL/min

B. 4800 mL/min

C. 5200 mL/min

✓D. 6000 mL/min

E. 6200 mL/min

Rationale: $VA = (VT - VD) \times F$

$VA = (680 - 180) \times 12$

$VA = 400 \times 12$

$VA = 6000 \text{ mL/min}$

The physiological dead space is a combination of the anatomical dead space and the alveolar dead space. In a healthy person the anatomical dead space approximates to the physiological dead space because the alveolar dead space is negligible. In this patient there is a difference of 30ml between the anatomical and the physiological dead space. This is the value is the alveolar dead space. Therefore, the value for the physiological dead space should be used to determine the most accurate calculation of the alveolar ventilation in the patient.

Question #: 30

A 2-day old male infant born prematurely at 29 weeks, is brought to the emergency department with respiratory distress. In reference to the attached chart, which of the following sets of effects is most likely associated with this patient?

✓A. A

B. B

C. C

D. D

E. E

Rationale: This question is asking about what can happen if the surfactants are deficient in lungs. The lungs cannot expand and collapse at focal locations called atelectasis. This results in respiratory distress. There is increased efforts of breathing characterized by use of accessory muscles of breathing (flaring of nostrils), low partial pressure of oxygen, cyanosis (due to increased reduced hemoglobin, increased surface tension (because of decreased surfactants, and decreased compliance of the lungs.

Attachment:

	A	B	C	D	E
Flaring of nostrils, Cyanosis	✓	✓		✓	
Low PO ₂	✓		✓		✓
Increased surfactant		✓	✓	✓	✓
Increased Surface tension	✓		✓	✓	
Increased compliance		✓			✓

Resp distress.jpg